24春线性代数期末（回忆版）

1. Multiple Choice. Only one choice is correct.

(1) Let A be an $m\times m$ real symmetric matrix. Which of the following assertions is false? ( )

(A) If v_1 and v_2 are two m-dimensional real column vectors which satisfy $Av_1=v_1$ and $Av_2=0$, then v_1 and v_2 are orthogonal.

(B) There exists a real invertible matrix P such that $P^{-1}AP$ is diagonal.

(C) A has m distinct eigenvalues.

(D) The sum of algebraic multiplicities of the distinct eigenvalues of A is m.

(2) Let A be an $m\times n$ real matrix and $U\Sigma V^T$ be a ~~single~~ singular value decomposition of A. Which of the following assertions is false? ( )

(A) The columns of U are eigenvectors of AA^T.

(B) The columns of V are eigenvectors of A^TA.

(C) The eigenvalues of AA^T and A^TA are real and positive.

(D) AA^T and A^TA have the same set of positive eigenvalues.

(3) Let A be an $n\times n$ real matrix. If for any $x\in R^n$, we have $x^TAx=0$, then ( )

(A) $|A|=0$.

(B) $A^T=-A$.

(C) $A=0$. Where 0 denotes the $n\times n$ zero matrix.

(D) the eigenvalues of A are all zero.

(4) Which of the following matrices is NOT diagonalizable? ( )

(A) $\begin{bmatrix}0&0&0\\1&0&0\\0&2&1\end{bmatrix}$ (B) $\begin{bmatrix}1&2&-1\\2&0&0\\-1&0&0\end{bmatrix}$ (C) $\begin{bmatrix}0&0&0\\0&0&0\\1&2&-1\end{bmatrix}$ (D) $\begin{bmatrix}0&0&0\\0&1&0\\0&1&2\end{bmatrix}$

(5) Let A be a 3×3 permutation matrix. Then $\det(-(A^T)^{2024})$ equals ( ).

(A) 0 (B) 2024 (C) 1 (D) -1.

2. Fill in the blanks.

(1) Suppose $A=\begin{bmatrix}a&b&-\frac{3}{7}\\-\frac{3}{7}&c&\frac{2}{7}\\\frac{2}{7}&d&e\end{bmatrix}$ is an orthogonal matrix. Then $a=$ _____, $e=$ _____.

(2) Let A and B be two $n\times n$ real matrices $|A|=4$, $|B|=3$ and $|A^{-1}+B|=2$, then $|A+B^{-1}|=$ _____.

(3) Let A be a 3×3 matrix. Suppose $|A|=-5$ and $A^2+4A-5I=0$, then the three eigenvalues of A are _____. Where I denotes the 3×3 identity matrix and

0 denotes the 3×3 zero matrix.

(4) Suppose $A=\begin{bmatrix}1 & a & 1\\ a & b & a\\ 1 & a & 1\end{bmatrix}$ and $B=\begin{bmatrix}2 & 0 & 0\\ 0 & b & 0\\ 0 & 0 & 0\end{bmatrix}$ are similar, then $a=$ _____.

3. Consider the following matrix $A=\begin{bmatrix}4 & 2 & 2\\ 2 & 4 & 2\\ 2 & 2 & 4\end{bmatrix}$

(a) Find all the eigenvalues of A.

(b) Find an orthogonal matrix Q such that $Q^{-1}AQ$ is diagonal.

(c) Compute A^k for any positive integer k.

4. Find the determinant of the following 7×7 matrix: $A=\begin{bmatrix}5 & 3 & 0 & 0 & 0 & 0 & 0\\ 2 & 5 & 3 & 0 & 0 & 0 & 0\\ 0 & 2 & 5 & 3 & 0 & 0 & 0\\ 0 & 0 & 2 & 5 & 3 & 0 & 0\\ 0 & 0 & 0 & 2 & 5 & 3 & 0\\ 0 & 0 & 0 & 0 & 2 & 5 & 3\\ 0 & 0 & 0 & 0 & 0 & 2 & 5\end{bmatrix}$

5. Let $A=\begin{bmatrix}1 & 1\\ 2 & 3\\ 4 & 5\end{bmatrix}$.

(a) Find all the eigenvalues and their corresponding linearly independent eigenvectors of $P=A(A^TA)^{-1}A^T$.

(b) Show that P is diagonalizable.

6. Consider the following quadratic form:

$$Q(x,y,z)=\lambda(x^2+y^2+z^2)+2xy+2xz-2yz$$

(a) For what values of λ is $Q(x,y,z)$ positive definite?

(b) For what values of λ is $Q(x,y,z)$ negative definite?

(c) Find the type of quadric surface defined by the following equation:

$$3(x^2+y^2+z^2)+2xy+2xz-2yz=1$$

7. Let A and B be $n\times n$ real matrices. The trace of A is defined to be the sum of all of its diagonal entries:

$$tr(A)=a_{11}+a_{22}+\cdots+a_{nn}.$$

(a) Suppose A is similar to B, prove that

$$tr(A)=tr(B)$$

(b) Let A and B be real symmetric positive semidefinite matrices. Show that $tr(AB)\geq0$.

(c) Suppose A and B are real symmetric positive semidefinite matrices and $tr(AB)=0$. Show that $AB=0$. Where 0 denotes the $n\times n$ zero matrix.